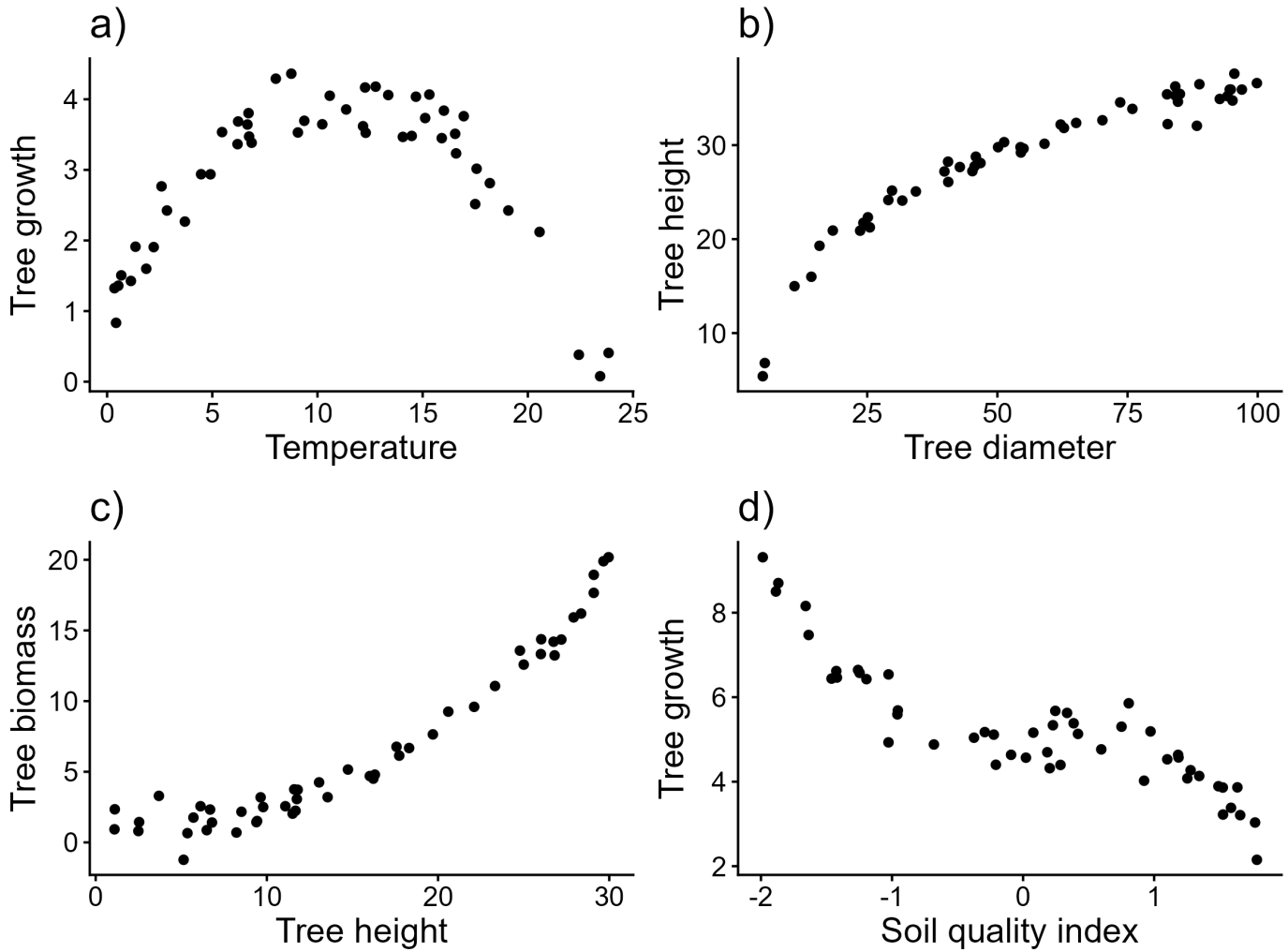
A gentle introduction to Generalized Additive Models

Lionel Hertzog

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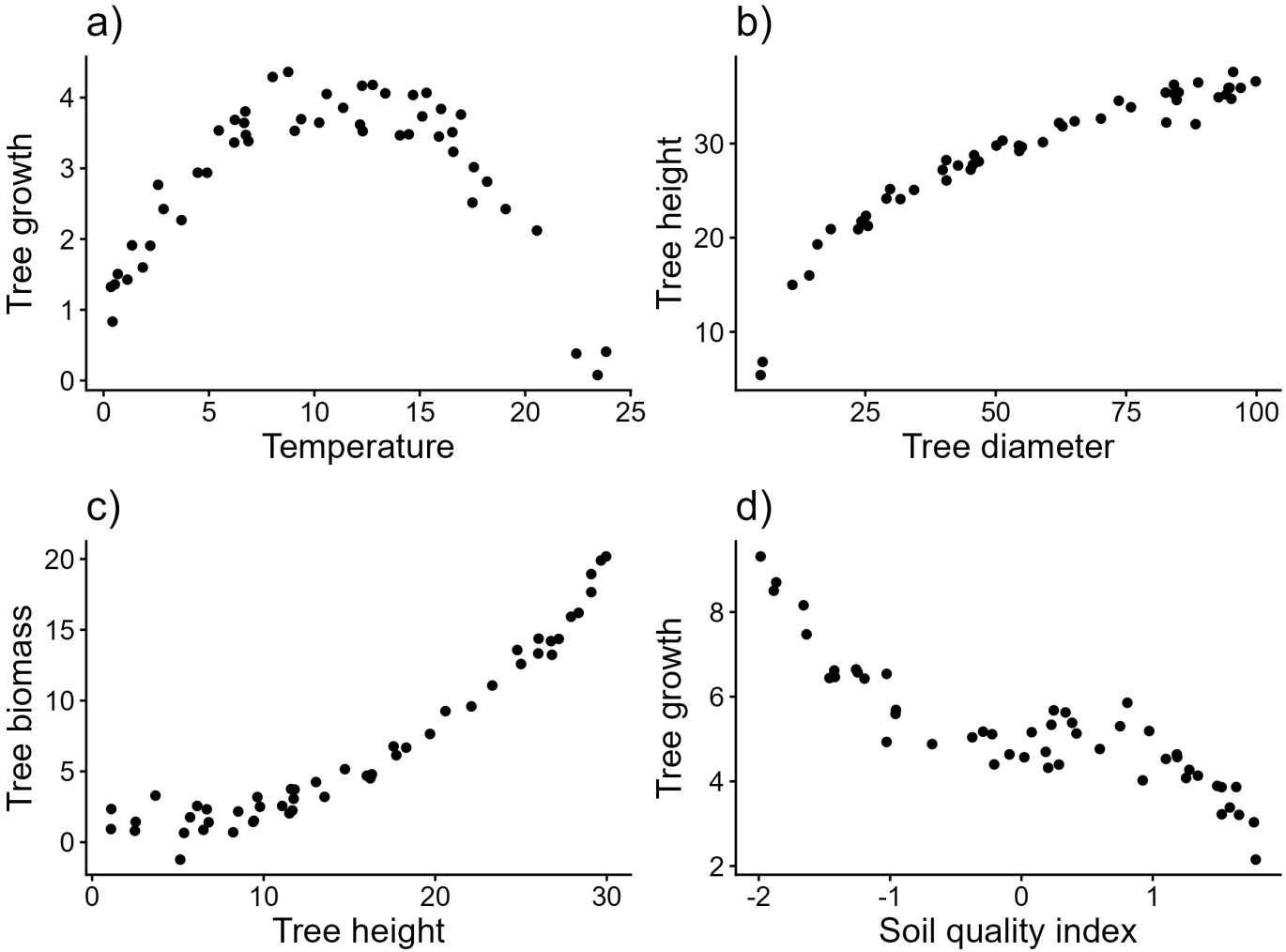
On non-linearity

In biological systems, such as forests, most relationships between variables are non-linear to some degree.



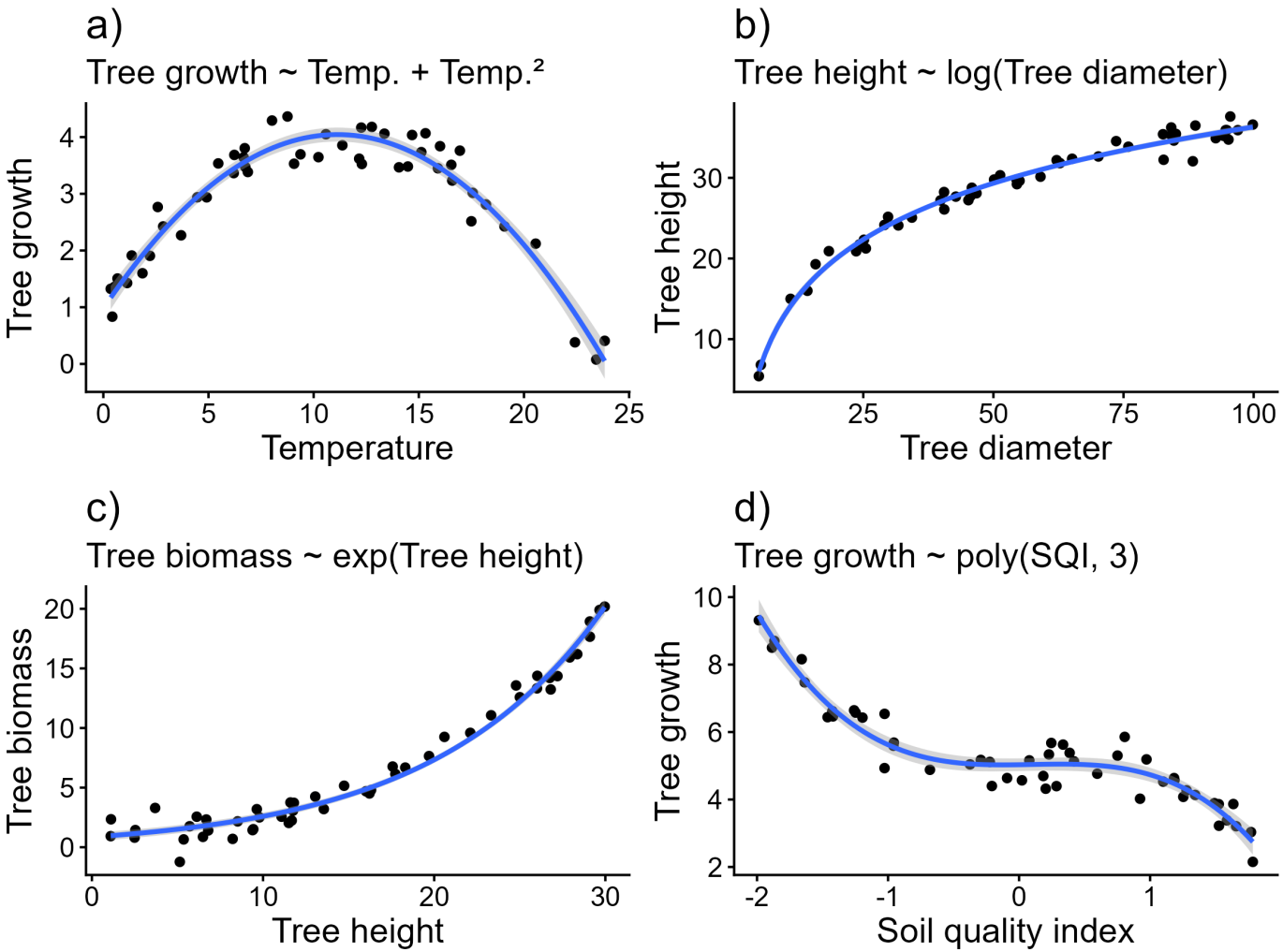
Linear models that are not so linear

Among the following 4 relationships, for which ones would a (G)LM be incorrect?



Linear models that are not so linear

All the relationships in this figure can be estimated with a (G)LM.



So - why bother?

- In cases of non-linear relationships, (G)LMs nevertheless show limitations
- Polynomial functions remain relatively "rigid"
- Polynomial functions quickly become unstable as the degree increases
- Adding a slightly odd observation can derail the estimation of a high-degree polynomial relationship (see practical session)
- Take home: if a (G)LM does the job, you might as well use it, but if the problem at hand does not give a satisfactory answer with basic transformations (sqrt, log ...) or polynomials of degree 3-4 at most, GAMs are an alternative.

Generalized Additive Models

GAMs are GLMs on steroids.
Reminder, a GLM is written as:

$$\begin{aligned} y &\sim EF(\mu, \cdots) \\ g(\mu) &= \alpha + \beta_1 * x_1 + \cdots \end{aligned} \tag{1}$$

μ is the expected value, $g()$ the link function and EF a distribution from the exponential family (normal, Poisson ...).
A GAM is written as:

$$\begin{aligned} y &\sim EF(\mu, \cdots) \\ g(\mu) &= \alpha + f(x_1) + \cdots \end{aligned} \tag{2}$$

f is a "smooth" function that can take a wide variety of shapes.
Take-away: The only difference between a GLM and a GAM is the use of smooth functions.

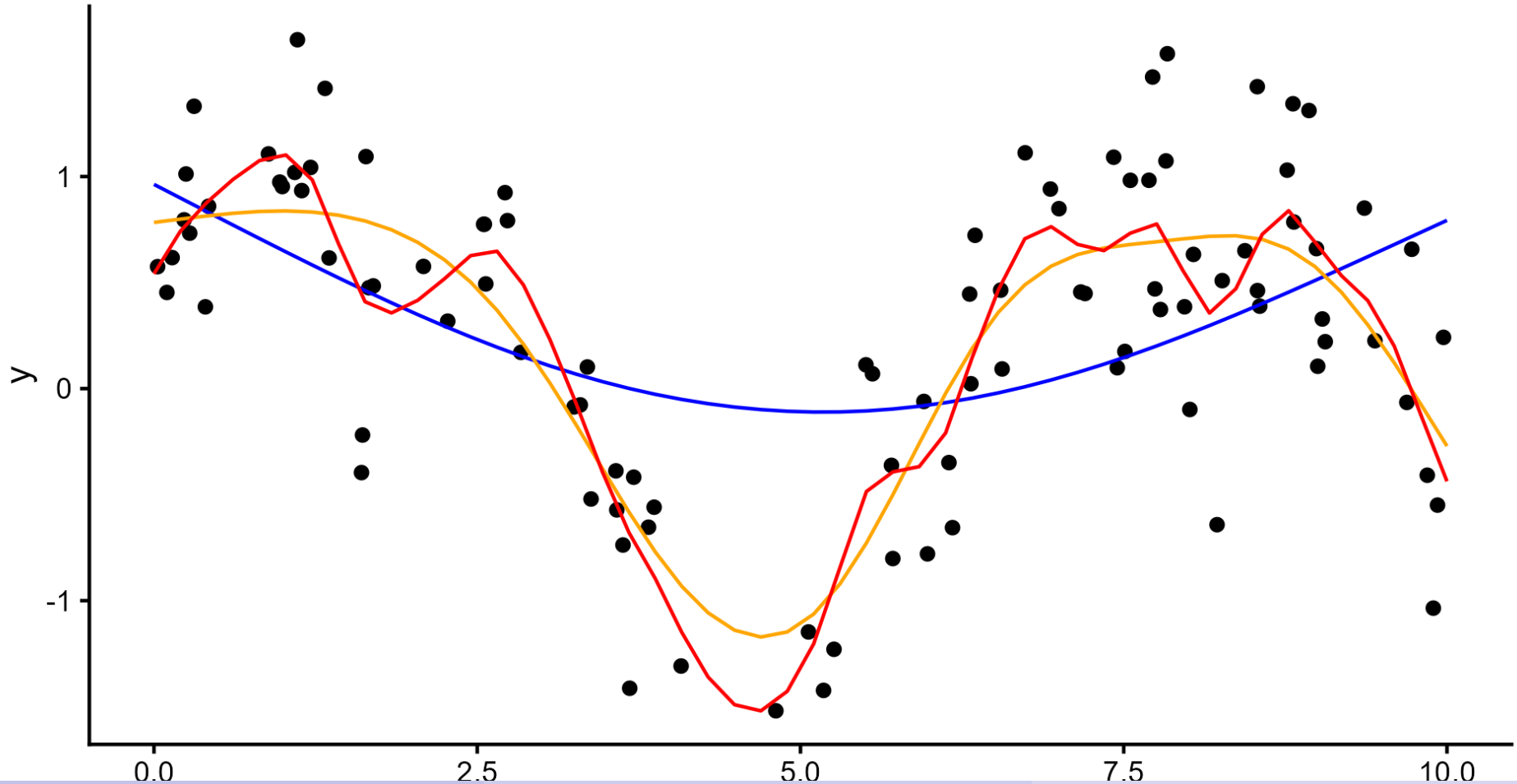
A bestiary of smooth functions

The *mgcv* library in R lets you choose among 14 smooth functions, see `?smooth.terms` for a list of the options.
Moreover, smooths can be combined across variables to create multivariate smooths using *tensors*.
In this course we will cover two smooth functions: thin plate regression splines (the default option in mgcv) and cyclic cubic regression splines. We will also cover 2D tensors.

Smooths

Objectives

The smooth function must fulfil 2 objectives: (i) fit the data well and (ii) not be too complex (wiggly)



Smooths - basis functions

Smooth functions are written as:

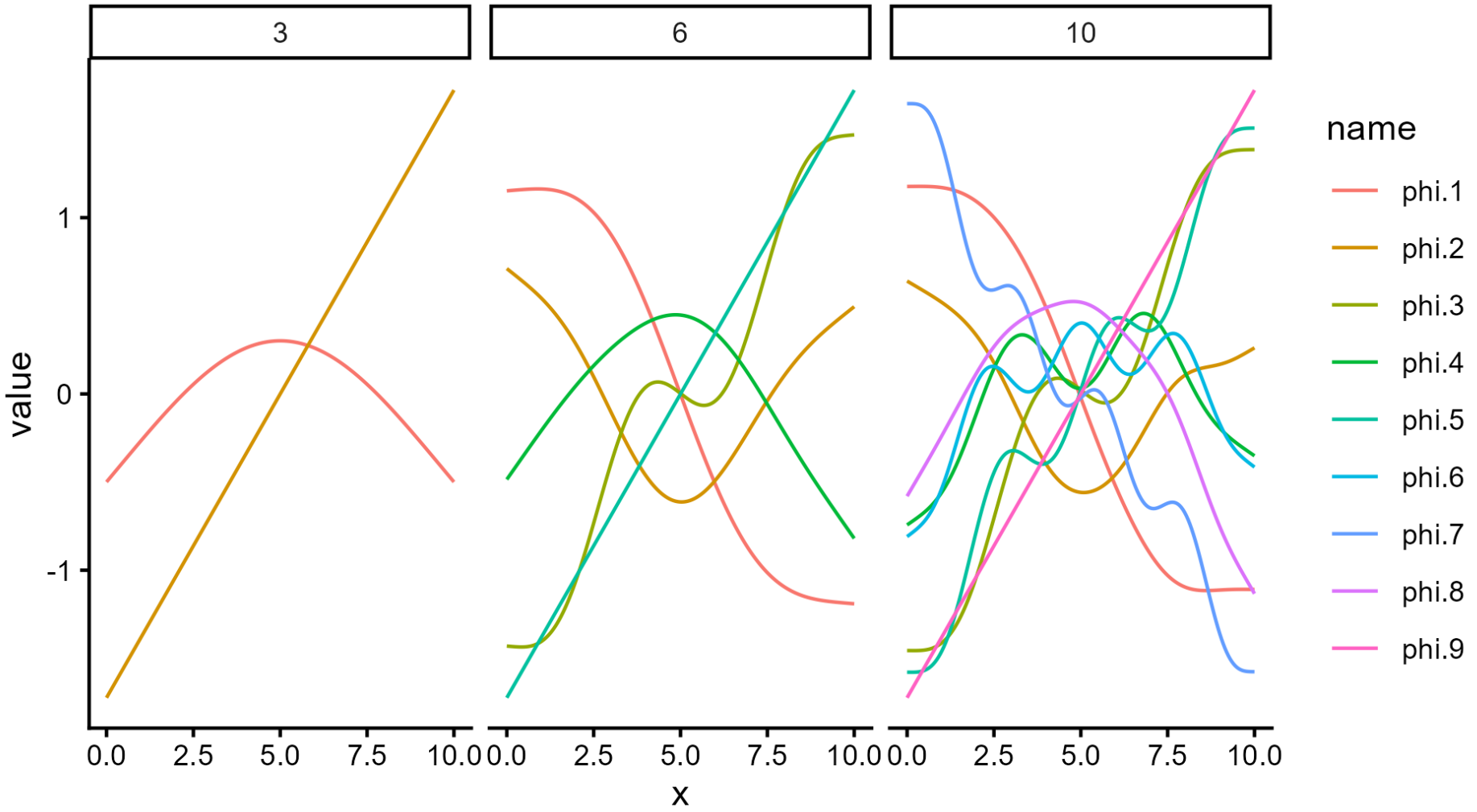
$$f(x) = \sum_{k=1}^{K} \beta_k * \phi_k(x) \tag{3}$$

With K the number of *basis functions*, β coefficients to be estimated as in a (G)LM and ϕ the *basis functions*.
Reminder: in a GLM with J covariates, the predictor (μ) can be written as:

$$\mu = \sum_{j=1}^{J} \beta_j * x_j \tag{4}$$

Basis functions

Basis functions are estimated in a **deterministic** way from the covariate matrix; they are the building blocks of the smooth functions.

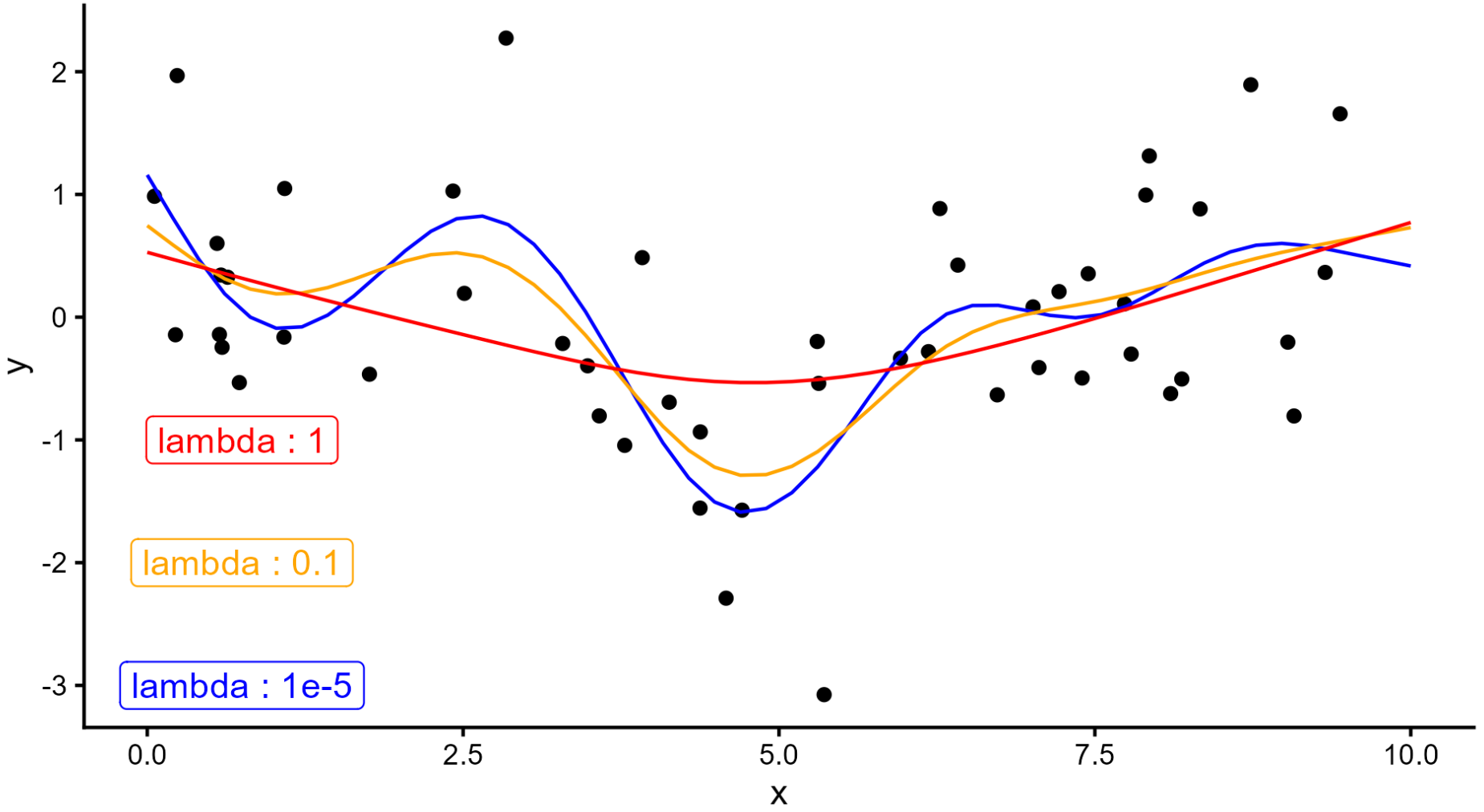


Basis functions, K and λ

The shape of the *basis functions* depends on the type of smooth chosen (e.g. thin plate regression spline or cubic spline, see practical session).
The number of *basis functions* controls the complexity of the smooth: the larger K is, the more complex the smooth can be. The exact value of K is not too important, but K must be large enough.
The degree of wiggliness is controlled by another parameter, λ, which is optimized with respect to the data when the model is fitted. λ can be seen as a penalty on smooths that overfit the data.

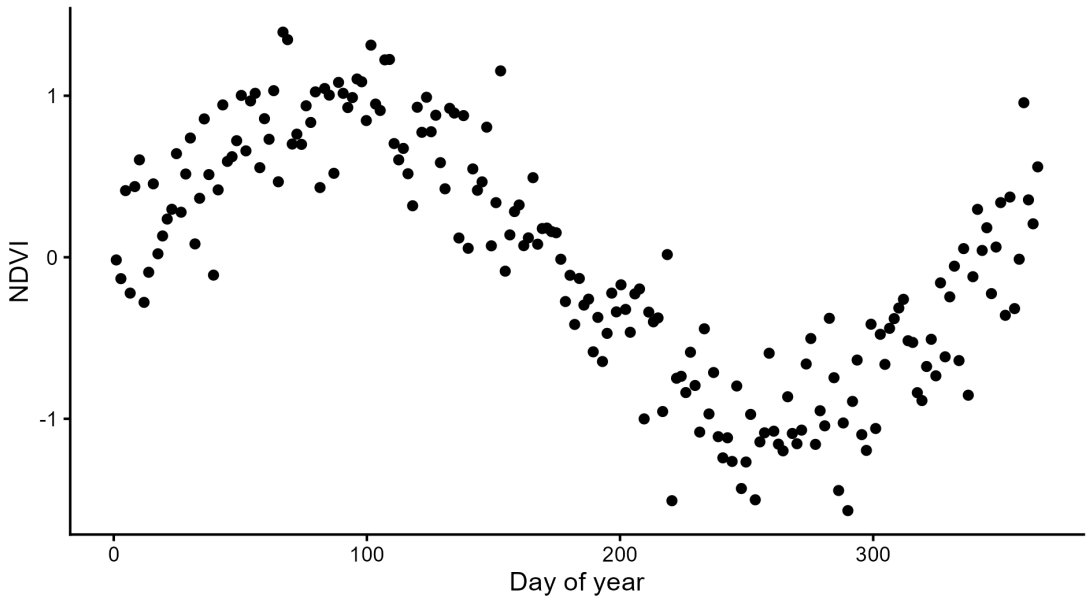
λ - just right

By default, the lambda parameter is optimized with respect to the data.



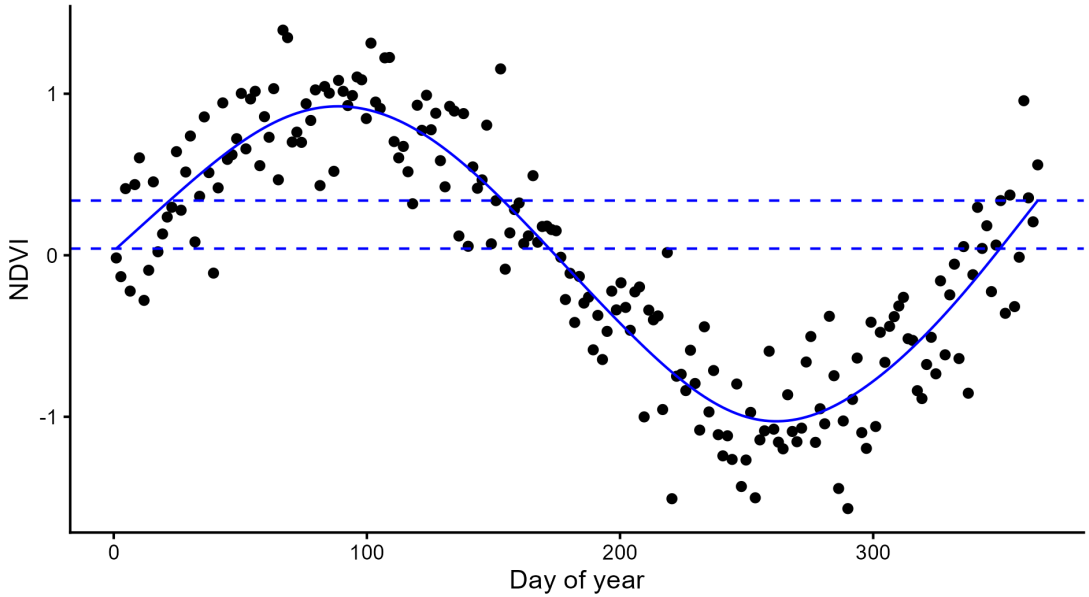
Cyclic cubic splines

- Some data are cyclic in nature, notably data collected throughout a year or around a compass
- With this type of data, there should be no discontinuity between the first and last values predicted by the model (smooth connection), e.g. between 00:00 and 23:59.



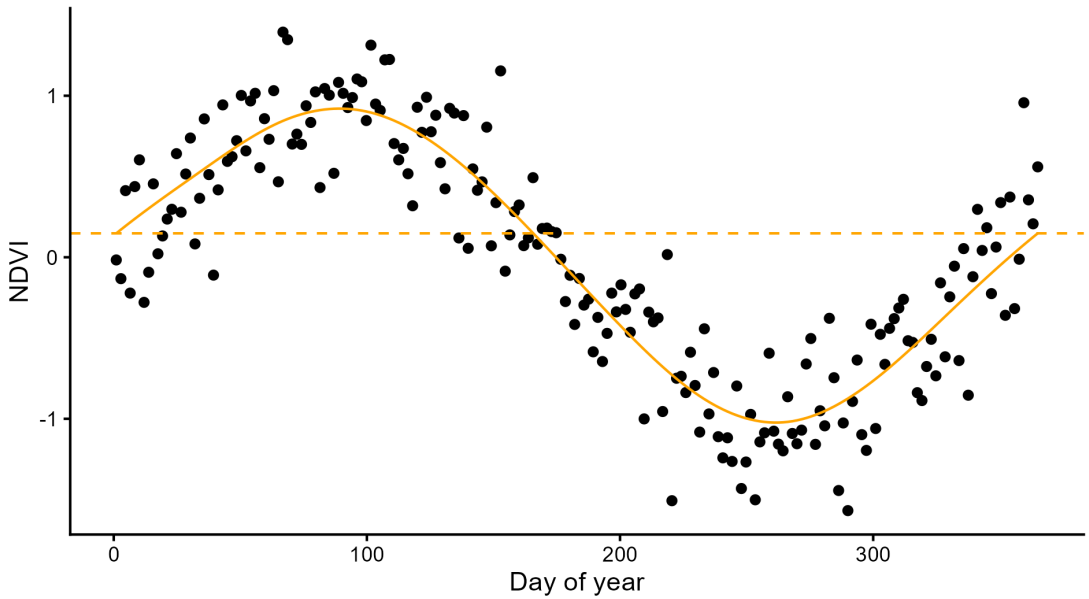
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- Some data are cyclic in nature, notably data collected throughout a year or around a compass
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- Classic smooths such as `tp` or `cr` do not have this property and generate discontinuities at the ends of cyclic data
- A class of smooths, the *cyclic cubic splines*, ensures continuity for this type of data.



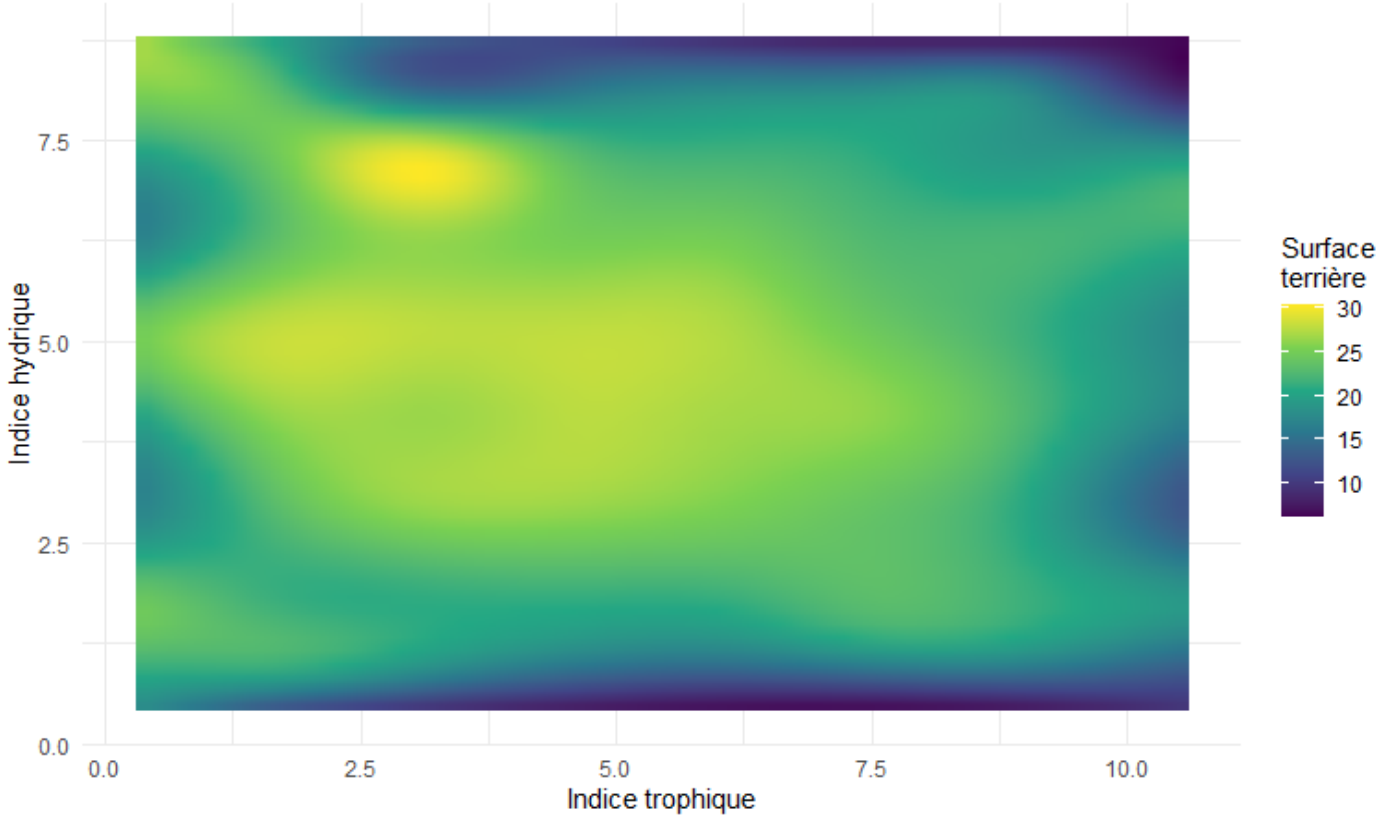
Cyclic cubic splines

Cyclic cubic splines are cubic splines that control three aspects at the endpoints **a** and **b**:

1. The value of the smooth is equal at both endpoints: $f(a) = f(b)$
2. The slope of the smooth is equal at both endpoints: $f'(a) = f'(b)$
3. The curvature of the smooth is equal at both endpoints: $f''(a) = f''(b)$

Multivariate splines

Smooths can be multivariate, over 2, 3, ... dimensions.



In practice, beyond two variables smooths become difficult to interpret.

Multivariate splines - tensor product

- Suitable when the variables are in different units (temperature, precipitation); the tensor optimizes one λ parameter per variable.
- By default, mgcv uses a *cubic splines* smooth, but other options can be used, and it is even possible to have a different type of smooth for each variable.
- Allows interactions between variables to be taken into account - a flexible response surface.

Tensor product

In detail, for a *tensor product* between two variables (x_1 and x_2), with K and L the number of *basis functions* used for each variable, the estimation is done as follows:

$$g(\mu) = \alpha + \sum_{k=1}^{K} \sum_{l=1}^{L} \beta_{kl} \phi_{1k}(x_1) \phi_{2l}(x_2) \tag{5}$$

In effect, each *basis function* of one variable is multiplied by each *basis function* of the other, and the products are weighted by β coefficients to be estimated.

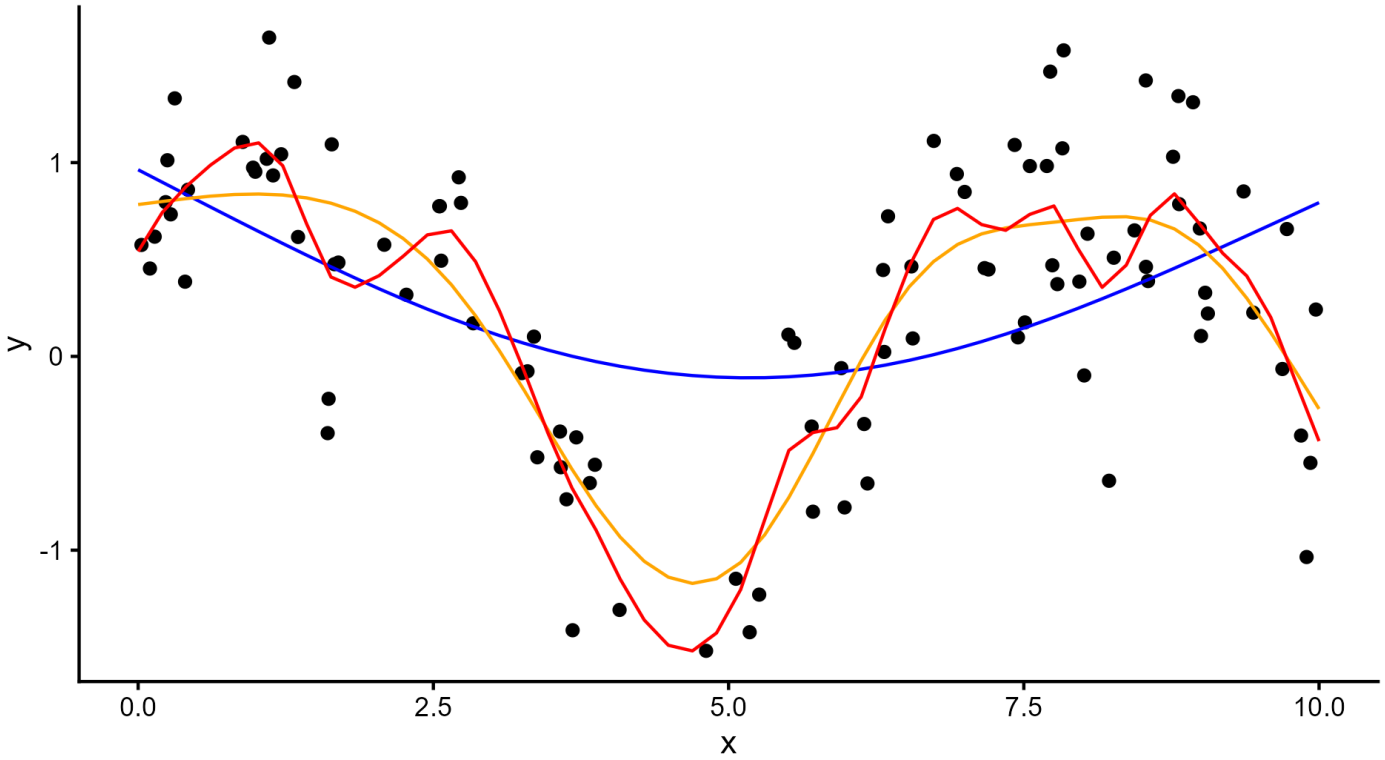
GAM - Model evaluation

Several aspects of a GAM must be evaluated in order to interpret the results reliably:

- Overfitting control
- Distribution of residuals
- Concurvity - collinearity for GAMs
- Adequacy of the number of *basis functions* used

Overfitting checks in GAM

Reminder: we want a model that follows the general trends in the data. A model that overfits will have reduced predictive ability.



Overfitting checks in GAM

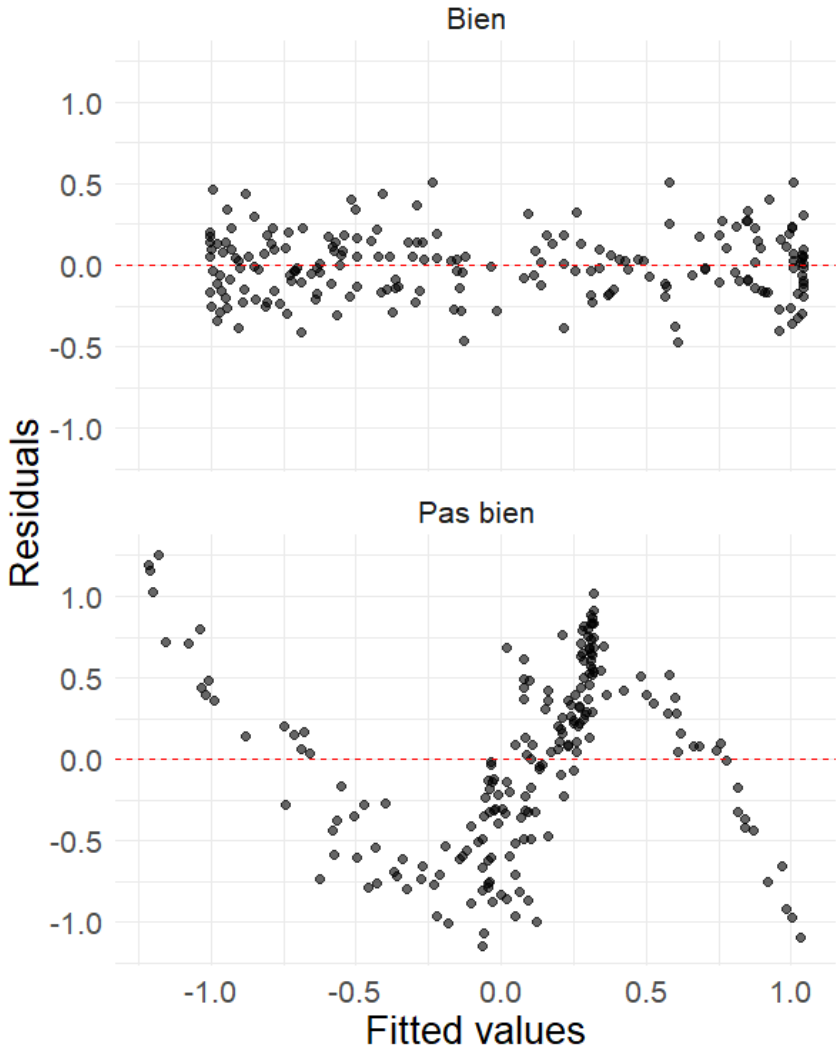
Reminder: we want a model that follows the general trends in the data. A model that overfits will have reduced predictive ability.
To assess the degree of overfitting, several methods exist:

- Visual exploration of the smooths (plot.gam)
- Effective degrees of freedom: if the value is close to k-1, there is a risk of overfitting
- Cross-validation: comparing model quality between training data and test data (not used to fit the model).

Distribution of residuals

Once the effects of the covariates have been accounted for, no pattern should remain in the residuals with respect to:

- The model predictions (μ)
- The covariates included in the model
- The covariates not included in the model



Concurvity

When several smooths are correlated, model estimation is more difficult - risk of non-convergence - and unstable - small variations in the values can have a disproportionate influence on the model.
The **concurvity** function of the `mgcv` package computes three concurvity indices - 0 means no sign of a problem, 1 means complete correlation between smooths.

k.check

If the number of *basis functions* (K) associated with a smooth is too low, the estimated smooth will not be optimal - not enough flexibility to fit the data.
The **k.check** function of the `mgcv` package computes an index of the adequacy of the K parameter. A value below 1 with a p-value < 0.05 indicates a problem.

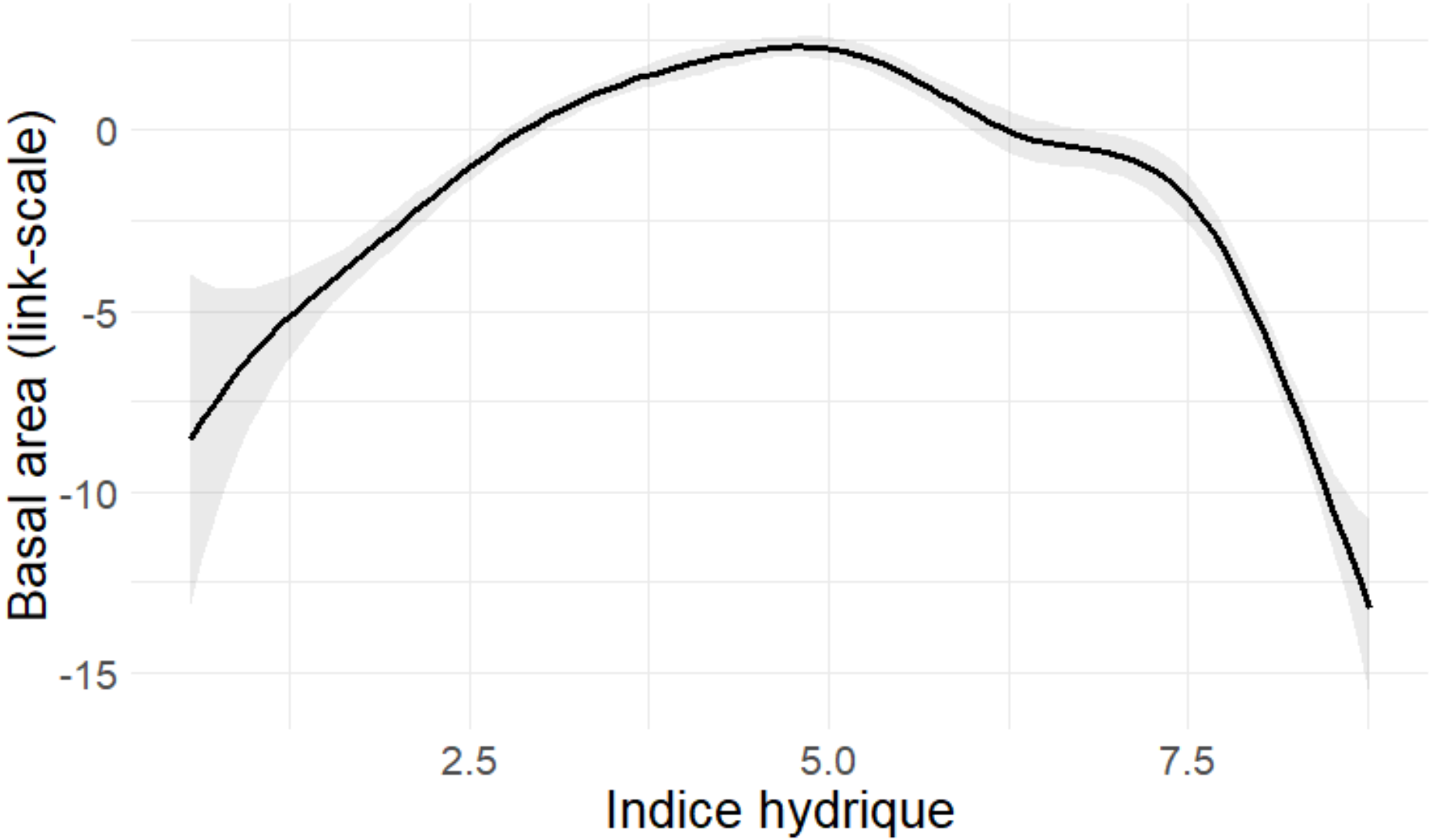
Interpreting a GAM

The coefficients estimated by the model are associated with the *basis functions*, so they have no biological interpretation:

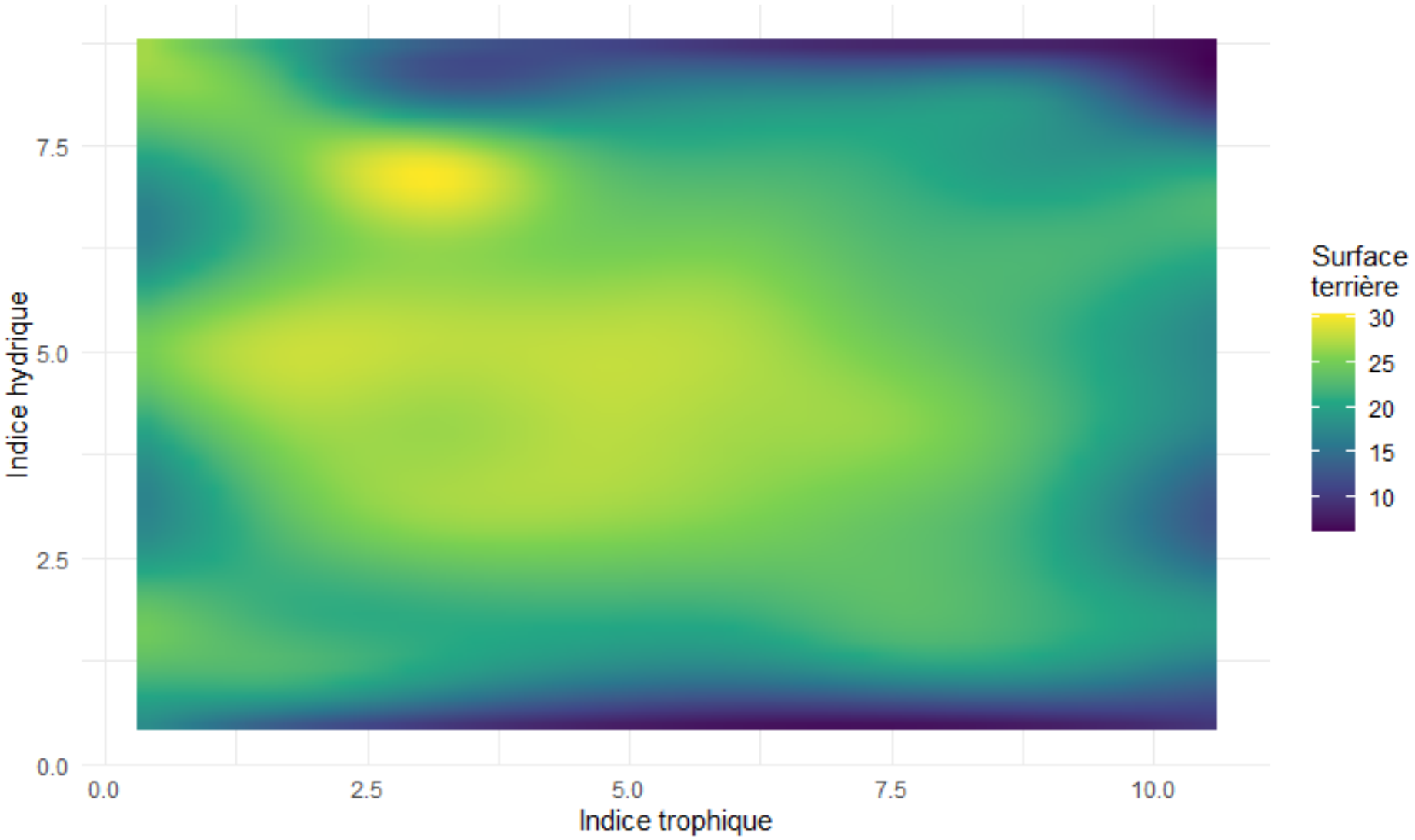
$$f(x) = \sum_{k=1}^{K} \beta_k * \phi_k(x) \tag{6}$$

So, unlike in a GLM, the coefficients are not interpretable. Only the curves (surfaces) estimated by the model can be interpreted from a biological point of view.

Interpreting a GAM



Interpreting a GAM



To take home - pros and cons

- GAMs make it possible to estimate flexible relationships between variables
- A large number of *smooths* are available in the mgcv package in R (e.g. *random effect smooths*)
- Because of the flexibility of the estimated relationships, there is a potential risk of the model overfitting the data
- The coefficients estimated by the model have no biological interpretation, and it can also be hard to interpret - and to link to scientific hypotheses - curves with many variations.

Time for Rrrr